

Breakdown voltage and its temperature coefficient for a cryogenically operated SiPM

A single-photoelectron charge calibration across three cooldowns

Internal characterisation note

July 2023

Abstract

We calibrate the single-photoelectron response of a silicon photomultiplier operated in a cryostat at three temperatures and extract its breakdown voltage at each, together with the temperature coefficient dV_{bd}/dT . The observable is the charge separation between the noise and first-photoelectron populations of the integrated-pulse spectrum, measured in ADC-ns, which is independent of the uncalibrated analogue gain of the readout chain. Breakdown voltage is obtained by extrapolating that separation against bias to zero, about 6 V below the lowest measured bias point. We obtain $V_{bd} = 43.037 \pm 0.052$, 43.972 ± 0.062 and 45.073 ± 0.027 V at measured temperatures -98.58 , -77.67 and -57.72 °C, and $dV_{bd}/dT = 50.4 \pm 2.2$ mV/K. The dominant systematic is not statistical: within each cooldown the bias ladder was stepped while the cryostat was still drifting, so bias and sensor temperature are correlated at $\rho > 0.98$ and a fit against bias alone absorbs part of the temperature dependence into the intercept. Correcting for that correlation moves V_{bd} by $+0.07$ to $+0.10$ V, comparable to the entire spread between defensible analysis methods, and is the single decision that most affects the result.

1 Motivation

A SiPM's gain is set by its over-voltage $\Delta V = V_{bias} - V_{bd}$, and V_{bd} moves with temperature. Any detector that operates SiPMs cold must therefore know $V_{bd}(T)$ before it can choose a bias set point, equalise gain across channels, or fix a counting threshold: everything downstream inherits that calibration. The measurement is routine in the sense that many groups perform it, and load-bearing in the sense that an error in it propagates everywhere. Published characterisations of cryogenic SiPMs describe the same procedure, at liquid-xenon temperature [1] and warm-and-cold as production QA [2]. This note is a bench measurement on a single device, carried out at three cryostat setpoints in one campaign of runs.

2 Data and setup

The device under test is read out on channel 3 of a CAEN DT5720 desktop digitizer (250 MS/s, 12-bit, 2 V_{pp} input range) through a cold preamplifier and a warm second stage. Two further channels carry the trigger companions and are not used here. The analogue chain was never calibrated against a known charge, so no conversion to electrons is attempted and every charge quantity in this note is quoted in ADC-ns.

Acquisition covers four cooldowns at controller setpoints -102 , -95 , -78 and -58 °C. At each setpoint a ladder of bias voltages was stepped in 1 V increments, and each bias point was recorded several times at different leading-edge trigger thresholds. In total 172 files were written, each holding 700 events of 1024 samples per channel. Cryostat temperatures were logged independently of the digitizer throughout.

Files are written in the DT5720 standard-firmware event format: four 32-bit header words carrying the event size, the channel mask, an event counter and a trigger time tag, followed by the sample payload with two 12-bit samples packed into each 32-bit word, the earlier sample in the low half. We established this layout from the manufacturer's documentation and then checked it against the files rather than accepting it: the record length divides every file exactly, event counters are contiguous, trigger time tags advance monotonically, and the event-averaged pulse is smooth. That last check is the one that matters. Per event, the minimum, maximum and sum of the samples are identical under a reversal of the packing order, so only the pulse shape distinguishes the two readings — and the reversed order produces a visibly serrated waveform whose mean absolute second difference is three times larger.

3 Method

3.1 Pulse charge

For each event we fit a low-order polynomial baseline through the sample range ahead of the pulse and integrate the baseline-subtracted waveform over a fixed gate. Figure 1 shows the event-averaged pulse; it peaks

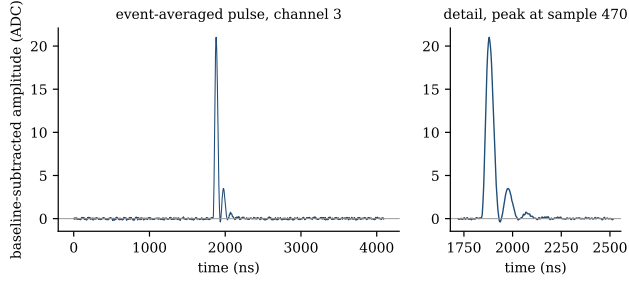


Figure 1: Event-averaged, baseline-subtracted pulse on the device under test, with the peak region on the right. The pulse is resolved on the 4 ns sampling grid and its shape is what distinguishes the correct sample-packing order from its reversal.

at sample 470, i.e. 1880 ns into the record, and the gate is placed around that peak. Block means rather than medians are used for the baseline: the samples are 12-bit integers with a baseline spread of a couple of counts, and a median snaps onto integer steps and shifts every area by a constant.

3.2 The observable

The spectrum of integrated charges is not a single peak. It carries a noise population from events in which the device under test did not fire, a first-photoelectron population, and higher orders that at these statistics hold too few events to locate a centre. We therefore measure the *separation* between the noise and first-photoelectron populations, not the absolute position of either.

This choice is not cosmetic. The noise population does not sit at zero: it carries whatever residual offset the baseline estimator leaves behind, and that offset is common to both populations, so it cancels in the difference and does not cancel in an absolute peak position. Reporting a peak position instead of a separation carries that offset into an extrapolation several volts long. During this analysis the two definitions were seen to disagree by of order 0.1 V in V_{bd} purely by measuring different quantities.

3.3 Acquisition thresholds

Each bias point was recorded at several trigger thresholds, and the threshold selects which events reach the file. That is a selection on pulse amplitude, and if it were also selecting on integrated charge it would bias the separation. Figure 2 shows the extracted separation against threshold at every bias point of all three calibrated cooldowns. It is flat within the fit errors at every setting; we therefore merge thresholds per bias point and carry the spread across them as a systematic rather than assuming it away.

	−102 °C	−78 °C	−58 °C
$\rho(V, T)$ within campaign	0.997	0.988	0.989
dT/dV (K/V)	0.26	0.24	0.21
ΔV_{bd} if uncorrected (V)	−0.071	−0.103	−0.101

Table 1: Within each cooldown the bias ladder was stepped while the cryostat was still settling, so bias and sensor temperature are almost perfectly correlated. A fit against bias alone therefore absorbs part of the temperature dependence and biases the intercept low.

3.4 Which cooldowns can be calibrated

Breakdown voltage is the intercept of separation against bias, so a cooldown needs a ladder of distinct bias voltages before it can carry one. Three of the four cooldowns do: −102 °C spans 49–54 V, −78 °C spans 50–55 V, and −58 °C spans 51–56 V, each in 1 V steps at three thresholds. The −95 °C cooldown sits at a single bias voltage. No number of events fixes a slope from one abscissa, so it cannot carry an independent breakdown voltage and we do not report one for it.

3.5 Temperature

The temperature used for each cooldown is the mean of the cryostat log over that cooldown’s acquisition window, with the standard deviation over the same window quoted alongside it. The controller setpoint is not used as a measurement: the setpoints and the logged means differ by up to 3.4 K, which is large enough to move the coefficient well outside its uncertainty.

4 The temperature–bias confound

The bias ladders were stepped upward while the cryostat had not fully settled. Within each cooldown, bias voltage and logged temperature are consequently correlated at $\rho > 0.98$ (Table 1), with the sensor moving 0.21–0.26 K per volt of bias.

This matters because the separation depends on both quantities. Fitting it against bias alone lets the fit absorb part of dQ/dT into the slope, which tilts the ladder and moves the intercept. Switching the correction off in our own pipeline moves V_{bd} by −0.071, −0.103 and −0.101 V at the three cooldowns — always low, and largest where the ladder is longest.

We therefore fit against a drift-corrected bias,

$$X_i = V_i - \beta (T_i - T_c), \quad (1)$$

where T_i is the logged temperature at bias point i , T_c the campaign temperature, and β the breakdown-voltage temperature coefficient itself. Because β is what we are

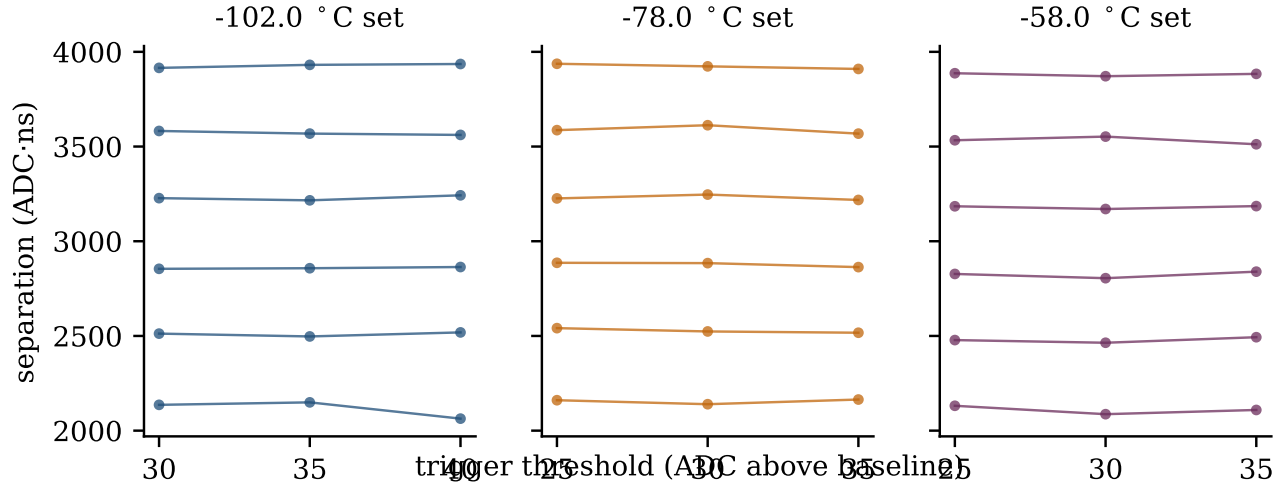


Figure 2: Extracted noise-to-1PE separation against acquisition threshold, one trace per bias point, for the three calibrated cooldowns. Flat within errors: the threshold selects events by amplitude but does not measurably select on the charge observable.

measuring, it is iterated to self-consistency across the three cooldowns rather than assumed. Four iterations suffice.

The size of this correction deserves emphasis: at 0.07–0.10 V it is comparable to the entire spread between defensible analysis pipelines applied to the same files. An analysis that ignores it is not wrong by a rounding error; it is wrong by about as much as the method choice itself can justify.

5 Results

Figure 3 shows the separation against bias at each cooldown with the fit and its residuals. The relation is linear across the measured range; reduced χ^2 values are 0.53, 0.64 and 0.10, and the residuals scatter rather than following a shape. We fit a straight line for that reason and note what the alternative would cost: the intercept lies about 6 V below the lowest measured bias point, so a one per cent error in the slope moves V_{bd} by roughly 0.085 V, and a quadratic through the same points would move it by 0.3–0.6 V. Curvature has to be established before it is followed, not inferred from residuals inside the measured range.

Table 2 collects the per-cooldown results. Figure 4 shows V_{bd} against measured temperature. A weighted straight-line fit gives

$$\frac{dV_{bd}}{dT} = 50.4 \pm 2.2 \text{ mV/K}, \quad (2)$$

with residuals consistent with the per-cooldown uncertainties. The measured slope of the separation against bias is stable across cooldowns to better than 1.3%,

Setpoint (°C)	−102	−78	−58
Measured T (°C)	−98.58	−77.67	−57.72
σ_T over window (K)	0.86	0.48	0.40
Slope (ADC·ns/V)	360.6	357.7	356.1
V_{bd} (V)	43.037	43.972	45.073
$\sigma_{V_{bd}}$ (V)	0.052	0.062	0.027
χ^2/dof	0.53	0.64	0.10

Table 2: Per-cooldown calibration. Temperatures are means over each cooldown’s acquisition window from the cryostat log, not controller setpoints.

as expected if the device’s over-voltage response is temperature-independent to first order over this range.

6 Uncertainties

Per-setting uncertainties come from the fit covariance of the population centres and from the spread across the acquisition thresholds at that bias point, whichever is larger. These propagate into the per-cooldown extrapolation and from there into the coefficient.

The systematics that matter here are correlated across cooldowns: a change of analysis choice moves all three breakdown voltages in the same direction. We therefore propagate them by recomputing the temperature slope under each variation rather than by inflating the per-cooldown errors and refitting, which would treat them as independent and overstate the result. Varying the integration gate across its plateau, the baseline model, and the estimator used to locate the two populations moves the coefficient by less than 1.5 mV/K in total. The drift correction of Section 4 is the exception

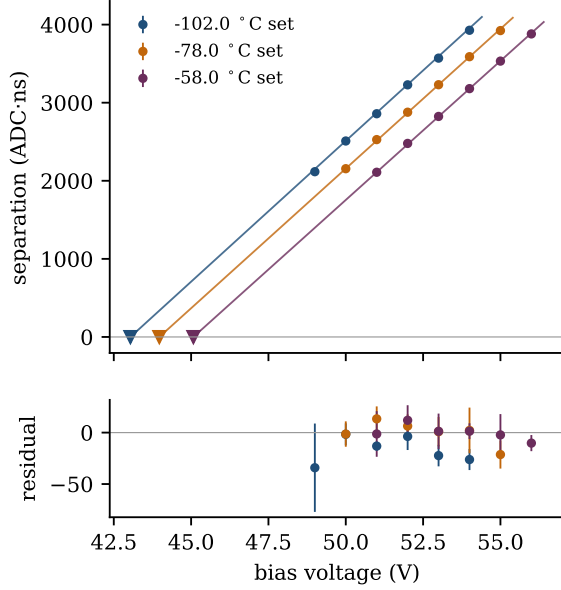


Figure 3: Noise-to-1PE separation against bias voltage for the three calibrated cooldowns, with the fitted line extrapolated to its intercept (triangles) and residuals below. The intercept sits about 6 V below the lowest measured point, so the slope carries a long lever arm onto V_{bd} .

and is not a variation but a correction: leaving it out is an error, not an alternative.

One limitation is worth stating plainly. The reference against which any such analysis can be checked is the spread of defensible methods applied to these same files, not an independently known truth. Two competent analyses of this dataset differ by up to 0.14 V in V_{bd} , and that number, rather than the fit uncertainty, is the honest scale on which a result should be judged.

7 Conclusion

The device’s breakdown voltage rises from 43.04 V at -98.6 °C to 45.07 V at -57.7 °C, giving $dV_{bd}/dT = 50.4 \pm 2.2$ mV/K over that range. The calibration rests on three decisions more than on any fitting detail: measuring a population *separation* rather than a peak position, taking temperatures from the cryostat log rather than the controller setpoint, and correcting the correlation between bias and temperature within each cooldown before extrapolating. The first two are matters of definition; the third is worth 0.07–0.10 V and is easy to miss, because a fit that ignores it looks entirely healthy.

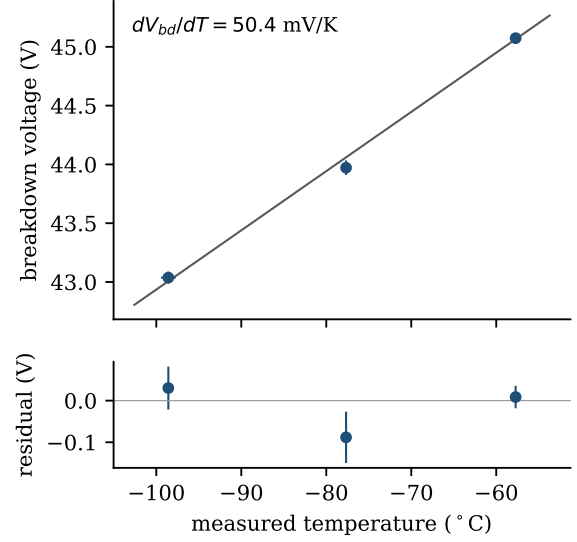


Figure 4: Breakdown voltage against measured cryostat temperature, with the weighted fit and residuals. Horizontal bars are the temperature spread over each acquisition window.

References

- [1] I. Ostrovskiy *et al.*, “Characterization of Silicon Photo-multipliers for nEXO,” *IEEE Trans. Nucl. Sci.* **62** (2015) 1825, [arXiv:1502.07837](#).
- [2] DarkSide-20k Collaboration, “Construction and characterisation of the DarkSide-20k veto silicon photo-multiplier tiles,” *Eur. Phys. J. C* (2026), [arXiv:2604.02551](#).
- [3] CAEN S.p.A., *DT5720 Desktop Waveform Digitizer — Technical Information Manual*, rev. 5.